IILM UNIVERSITY, GREATER NOIDA

DATA SCIENCE MASTER VIRTUAL INTERNSHIP

Industry-oriented learning in Data Engineering, Machine Learning and AI Applications

Student	GUFRAN ALI
Enrollment / Student ID	Engineering/2024/20837
Program & Semester	B.Tech CSE • Semester V
Organization	Siemens
Delivery Platform	EduSkills Foundation
Internship Period	April – June 2026 • 8 Weeks

Supported by Siemens • AICTE / National Internship Portal ecosystem • EduSkills

Internship at a Glance

- Eight-week virtual internship completed during April–June 2026.
- Focus: practical understanding of the data science lifecycle from data preparation to model evaluation.
- Learning environment: structured online content, assessments and visual data-science workflows.
- Primary industry support: Siemens; internship delivery platform: EduSkills Foundation.
- Completion certificate records Grade O (Outstanding), within the 90–100 range shown on the certificate.

Organization & Delivery Ecosystem

IILM University

Academic setting
B.Tech CSE
Greater Noida

EduSkills Foundation

Virtual internship delivery
Industry-oriented skilling
Academia–industry bridge

Siemens

Industry support
Technology, digitalisation
and data-driven transformation

The certificate supplied with the report records the participation of the Ministry of Education, AICTE, National Internship Portal, EduSkills and Siemens.

Problem Statement

- The main challenge addressed by the internship was the gap between learning isolated data-science concepts and applying them as one connected workflow.
- A useful analytical result depends on data quality, preparation, feature selection, model choice and validation—not only on the algorithm.
- The internship therefore emphasised a repeatable process for moving from a dataset to an interpretable analytical or predictive result.
- This approach is especially relevant to a B.Tech CSE student because it connects programming, databases, statistics, machine learning and software workflow design.

Objectives & Scope

- Understand the end-to-end data science lifecycle.
- Practise data inspection, filtering, transformation, selection and aggregation.
- Build familiarity with classification, regression, feature engineering and model validation.
- Interpret model outputs and compare alternative analytical approaches.
- Document workflows clearly enough to reproduce or explain them.
- Keep the work within an academic / training scope; no proprietary Siemens production data is claimed.

Technologies & Learning Areas

Python

Programming and data-science concepts

RapidMiner / Altair AI Studio

Visual workflow design and applied analytics

Data Preparation

Filtering, transformation, selection, aggregation

Machine Learning

Classification, regression, feature engineering

Validation & Evaluation

Cross-validation, performance views, model comparison

Visualisation

Patterns, trends and interpretation

System Architecture

Practical Data Science Architecture

Data Sources	Structured tables • transaction records • example datasets
Data Preparation	Attribute selection • filtering • transformation • missing-value handling
Analysis & Modelling	EDA • classification • regression • feature engineering
Validation & Evaluation	Cross-validation • performance measures • model comparison
Interpretation & Delivery	Findings • visual summaries • documented workflow

Methodology

1. **Understand** — Define the question and expected output
2. **Inspect** — Review attributes, types and data quality
3. **Prepare** — Filter, transform, select and reshape
4. **Model** — Apply an appropriate analytical method
5. **Validate** — Measure and compare model behaviour
6. **Interpret** — Explain findings and limitations

Practical Learning Highlights

- Visual data-processing exercises: attribute selection, filtering, transformation and aggregation.
- Model-building activities covering classification and regression workflows.
- Validation and comparison activities using performance measures and model-result views.
- Exposure to feature handling and the effect of preparation choices on downstream analysis.
- Process-organisation exercises that emphasised readable, modular analytical workflows.
- These activities were treated as training exercises rather than as claims about confidential industrial systems.

Outcomes & Skills Developed

Data handling	More confident with the preparation stage before modelling.
ML workflow	Clearer understanding of training, validation and evaluation.
Analytical reasoning	Better habit of checking whether outputs answer the original question.
Workflow thinking	Understands data science as a sequence of connected operations.
Technical communication	Able to document a workflow and explain its purpose.

Completion Certificate


शिक्षा मंत्रालय
MINISTRY OF EDUCATION
NATIONAL INTERNSHIP PORTAL
EduSkills
Nation Building Through Skills
AICTE - EduSkills 2026 VIRTUAL INTERNSHIP

Certificate of Virtual Internship

This is to certify that

Gufran Ali

IILM University, Greater Noida

has successfully completed the 8-weeks

Data Science Master Virtual Internship

During April - June 2026

May this Internship learning propel you toward a bright and successful career.

Supported By

SIEMENS

Gaurav Pai Head - Strategy and Operations Siemens Industry Software (India) Pvt. Ltd	Dr. Buddha Chandrasekhar Chief Coordinating Officer (CCO) AICTE, Ministry of Education	Shubhajit Jagadev Chief Executive Officer (CEO) EduSkills

Certificate ID : 43cd904240baf5aad33c

Student ID: STU699804e35f4f81771570403

INTERNSHIP GRADE O

GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

Completion evidence • 8-week Data Science Master Virtual Internship • April–June 2026 • Grade O (Outstanding)

Conclusion

The internship converted data-science concepts into a connected practical workflow.

- It strengthened the link between data preparation, modelling and evaluation.
- It provided exposure to industry-aligned visual analytics and machine-learning workflows.
- The experience complements the B.Tech CSE curriculum and provides a foundation for deeper work in machine learning and data-driven software systems.

Thank you

Selected References

- IILM University — Internship Report Format updated 26–27.
- Internship completion certificate — Data Science Master Virtual Internship, April–June 2026.
- EduSkills Foundation — About / Virtual Internship ecosystem.
- Siemens — India business profile and Digital Industries.
- Altair RapidMiner Academy / AI Studio documentation.

Detailed bibliography is included in the accompanying report.